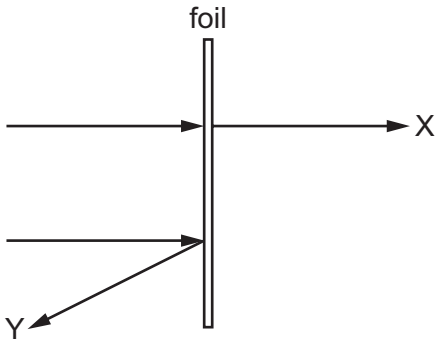
38 When a beam of α-particles is incident on a thin metal foil, most of them follow a path represented by path X in the diagram. A small number of α-particles follow a path represented by path Y in the diagram.



Which row correctly describes a conclusion that can be drawn from each of these observations about the structure of the atom?

	most follow path X	some follow path Y
A	atom is mostly empty space	atom contains something that repels α-particles
B	atom is mostly empty space	nucleus contains protons and neutrons
C	atom is neutral	atom contains something that repels α-particles
D	atom is neutral	nucleus contains protons and neutrons